Supplementary Material: Choosing a Fast Stand-In for a Slow Simulation

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S1. Proofs

Both results are standard; we include short proofs for completeness.

Proof of Proposition 1.. Let f^{-i} minimise $\sum_{j\neq i}(y_j - f_j)^2 + \lambda N\|f\|_K^2$ and let y' equal y except $y'_i = f_i^{-i}$. For any f, the objective with data y' equals the leave-one-out objective plus $(y'_i - f_i)^2 \geq 0$, so f^{-i}, which makes the extra term zero and minimises the first, also minimises the objective with data y'. By uniqueness of the ridge solution, $f^{-i} = Hy'$. Hence $f_i^{-i} = (Hy')_i = \hat{y}_i - H_{ii}y_i + H_{ii}f_i^{-i}$, so $(1-H_{ii})f_i^{-i} = \hat{y}_i - H_{ii}y_i$ and $y_i - f_i^{-i} = (y_i - \hat{y}_i)/(1-H_{ii})$. For $K = Q\,\mathrm{diag}(w)Q^\top$ with Q orthogonal, $H = Q\,\mathrm{diag}(w/(w+\lambda N))Q^\top$ and, since $\sum_j Q_{ij}^2 = 1$, $1 - H_{ii} = \sum_j Q_{ij}^2\,\lambda N/(w_j + \lambda N)$. For the polynomial model with a thin design matrix $F = U\,\mathrm{diag}(s)V^\top$, $1 - H_{ii} = (1 - \sum_j U_{ij}^2) + \sum_j U_{ij}^2 \lambda N/(s_j^2 + \lambda N)$. A bug in an early version of the tool (Sec. S4, V9) came from computing $1 - H_{ii}$ by subtraction and clipping it. □

Proof of Proposition 2.. Let $P = V_r V_r^\top$. For $x \in \bar{\mathbf{u}} + \mathrm{span}(V_r)$, $x - \mathbf{u} = (x - \bar{\mathbf{u}} - P(\mathbf{u} - \bar{\mathbf{u}})) - (I - P)(\mathbf{u} - \bar{\mathbf{u}})$, where the first term lies in $\mathrm{span}(V_r)$ and the second in its orthogonal complement. By Pythagoras, $\|x - \mathbf{u}\|_2^2 \geq \|(I - P)(\mathbf{u} - \bar{\mathbf{u}})\|_2^2$. Dividing by $\|\mathbf{u}\|_2$ and averaging over runs gives the claim. □

S2. Pre-specification record

Hypotheses, success criteria and analysis code were committed to the project's git repository before the results they concern (a self-hosted history, not an independent registry). Table S1 lists, for each experiment, the protocol commit and the first commit that contains its results; the check is automatic (`audit/prereg_order_check.py`) and shows that no protocol was committed after its results. It shows protocol-before-results in history, not

protocol-before-first-run. Experiments H, G3 and the Stage-6 experiments T, C and S are labelled post hoc in motivation: they were designed in response to earlier results or to reviews (Sec. S4).

Table S1: Order of pre-specification and results commits.

Exp	Protocol commit	First results commit	Hours before	Order ok
A	bcf7f30	2d0e2d4	9.33	yes
D	bcf7f30	2abe263	10.99	yes
B	bcf7f30	72cb02c	11.07	yes
A2	72cb02c	0bcb608	1.49	yes
E	72cb02c	703ad59	2.38	yes
A3	c9a7b93	82dbbe2	1.65	yes
G	c9a7b93	39b1b7f	3.66	yes
F	c9a7b93	39b1b7f	3.66	yes
H8	48a0698	84f2a21	0.08	yes
H32	48a0698	84f2a21	0.08	yes
G3	4cd411a	b37dd4e	0.05	yes
6T	a99c2c6	42fdacd	7.87	yes
6C	a99c2c6	1dc799e	53.43	yes
6S	a99c2c6	9ebaf04	56.97	yes
R	2619b07	a88ad5f	19.76	yes

S3. Earlier-stage hypotheses and verdicts

Table S2 lists the hypotheses and primary tests of the earlier stages (original 16 problems; independent replication on eight new problems; mechanism controls; longer-training and precision checks). Most confirmatory tests of the first stage were not supported; the one supported at cell level (unequal tuning) rests on 5 of 16 problems.

Pilot diagnostic. A frozen rule built from the original problems was tested on eight new problems from four new families: ρ =-0.38 (exact $p = 0.18$); out-of-sample gain 0.60 with Holm-adjusted $p = 0.051$ (10^6 bootstrap draws); the threshold rule classified 6 of 24 non-tie cells correctly. The pilot statistic also correlates with the number of inputs, and all problems with $f_1 < -2$ among the 24 have three inputs; we conclude that the diagnostic is inconclusive and do not use it in the tool.

Table S2: Earlier-stage tests. $\bar{y}$ as in (3).

ID	Criterion	Observed	p	Verdict
H1a (v1)	median $\bar{y}$ at N=32 positive (reduced basis better)	median -0.059	0.874	not supported
H1b (v1)	$\bar{y}$ falls with N in most problems	5/16 problems	0.962	not supported
H2 (v1)	pilot features beat $\{\log_2 N, p\}$	CI [-0.07, 6.40]	0.145 (Holm)	not supported
H4a (v1)	unequal tuning moves wins toward the more tuned side	10 vs 1 discordant cells	0.006	supported (cell level)
P1 (Exp A)	Spearman($f_1, \bar{y}$)< 0, 8 new problems	$\rho = -0.38$	0.180	not rejected
P2 (Exp A)	frozen rule beats baseline out of sample	gain 0.60	0.051 (Holm)	not rejected (knife-edge)
Q1/Q2 (A2)	≥ 4 of 7 smooth candidates with $\bar{y} \geq 1$; none network-clear	7/7; 0	–	met (confounded with p)
D1	narrow-range kernel ridge within 3× of GP in $\geq 75\%$ of cells	median ratio 14.3; 13% within 3×	–	rejected
E	iso-GP vs ARD-GP and kernel ridge	ratios 1.92, 8.1	–	marginal-likelihood rule met (low p only)
G1	full-field GP within 3× of POD+GP	median ratio 0.95	–	supported
H (post hoc)	wide-range kernel ridge within 3× (B=8 or 32)	medians 6.5, 2.7	–	neither rule met (mixed)
B1/B2	4× epochs: sign kept in $\geq 85\%$; median shift < 0.5	88%; 0.24	–	met, but 15 of 48 classes change
F	float64 + L-BFGS: ≥ 5 of 7 decisive gaps remain ≥ 1 decade	5 of 7	–	met

Robustness. Class labels agree between halves of the random-search draws in 83% of cells (sign agreement when both clear: 99%); 80% of reduced-basis-clear cells and 57% of network-clear cells have all three seeds beyond the tie band; class labels computed from the median instead of the mean error agree in 95% of cells; the extrapolation test set gives a median $\bar{y}$ of -0.03 against 0.16 on the interpolation set at $N = 1024$.

S4. Deviations from the plan

Every change to the plan is recorded with its date and reason in the repository (`research_design/deviations.md` for the first stage and `deviations_v2.md` afterwards); the text is reproduced here.

First stage (original 16 problems)

D1 Run restarts (operational). First launch killed (BLAS thread oversubscription) before any job finished; the container restarted twice later and killed in-flight jobs. Completed jobs are atomic (JSON written only on completion); the runner resumes and skips them. Identical code/protocol. *Effect:* none

D2 Interim analysis (disclosure). A premature analysis was run when 47 of 48 jobs existed (seeds 0,1 complete; my file count wrongly included `environment.json`). It used only complete seed batches (0,1) and was **not acted upon**; no code, hypothesis or setting changed. Output kept in `results/interim_2seed_disclosed/`. The confirmatory analysis is the 3-seed run. Verdicts of H1a, H1b, H2, H4a were identical in interim and final. *Effect:* none

D3 Seeds (protocol). Seeds 0–2 (mandatory) completed; extension seeds 3–4 **not run** (compute/time). *Effect:* declared in protocol

D4 Post-hoc exploratory analyses (exploratory). Added after seeing confirmatory results; every one is labelled *exploratory* in the manuscript and none changes a confirmatory verdict. See `results/exploratory/` . *Effect:* none

D5 Non-reproducible search sampling (bug) (disclosure + fix). Found in the fresh-clone audit: `hpo.py` seeded the random-search RNG with Python's per-process-salted `hash(method)`, so trials 1-7 of every hyperparameter search are sampled differently in every process. The experiment ran with this code. Effect: a from-scratch re-execution samples different trial configurations than those recorded (statistically equivalent, not bit-identical). All sampled configs and their validation/test errors are stored in `results/raw/*.json`; `code/experiments/replay_check.py` refits stored best configs and reproduces 20/20 recorded test errors exactly (relative diff 0.0). Fixed afterwards with `zlib.crc32` (regression test added). Results were **not** regenerated. *Effect:* none on reported numbers; re-runs will differ numerically

Later stages (replication, controls, tool evaluation)

V1 Exp B shift sign convention (bug found by independent audit (F1), fixed **before Exp B was analysed**). `shift = y6000 − y1500` is negative when NL improves; B2 "shift toward NL < 0.5" would have passed trivially. Replaced by `shift_toward_NL = y1500 − y6000`; unit test added. *Effect:* none (B not yet analysed)

V2 A2 baseline p0 miscounted (factual error in addendum v2.1 found by independent audit, verified). p0 stated 2/20; correct 3/20 (burgers_L1 f1 $= -1.9972$, $\bar{y} = 2.25$). Amended in addendum v2.2 **before A2 ran**: p0 $= 0.15$, success k ≥ 4 of 7. *Effect:* none (A2 not yet run)

V3 Diagnostic verdict wording (re-characterisation). "Not replicated" $\rightarrow$ *pre-registered criteria not met; inconclusive; carried by one family*

(nlheat 88.3 % of P2 gain; ρ = +0.49 without it); confounded with p (Spearman 0.708). Effect: none (numbers unchanged)

V4 Stage chronology (disclosure). Stage 3 (addendum v2.1) was committed within minutes after the Exp A and Exp D results; Stage 4 (addendum v2.2) after two independent audits. All preregistrations are on a self-hosted git history; git tags could not be pushed to the remote. *Effect:* none

V5 Shared search draws (design property found by audit (F2/F3)). The random-search RNG depends only on (seed, method): the same 7 sampled configs are used for every instance and N within a seed (3 distinct sets per method in the whole study). v1 additionally used a process-salted hash (D5). *Effect:* none on verdicts; changes how "tuning" must be described; sensitivity analysis planned

V6 Neural batch/step count (design property found by audit (F02)). batch = N for N $\leq$ 512 (1500 steps) but 512 for N = 1024 (3000 steps): the neural step count doubles at N = 1024. Exp F removes this (full batch). *Effect:* none on verdicts

V7 Test/doc fixes (housekeeping). stale test count fixed (36 instances); weaker leakage tests replaced; BLAS thread env set at the top of entry scripts. *Effect:* none

V9 SurroSelect smoke-test bug (before any registered Stage-6 run) — 2026-11-02. During a smoke test of the tool on the new problem `lshape_L1` (seed 0, N = 64 and 256) and a quick check of the KRR and polynomial members on `heatcomp_L2` and `convdiff_L1` (N = 64, 256), the wide-range kernel-ridge member returned a test error of 0.5 (10x worse than any other member). Cause: the closed-form leave-one-out residual divides by 1 - H_ii, which was clipped at 1e-6 (and computed by cancellation), producing spuriously tiny LOO scores for near-interpolating fits; the length-scale grid was also defined for inputs on [0,1] but with the [-1,1] limits of the benchmark's wide-range control. Fix (committed before any registered run): 1 - H_ii computed without cancellation (KRR: sum_j Q_ij^2 lam N/(w_j + lam N); polynomial: (1 - sum U_ij^2) + sum U_ij^2 lam N/(s_j^2 + lam N)), no clipping above 1e-12, ridge values with condition number above 1e12 skipped, length-scale grid [0.05, 15]·sqrt(p) on [0,1]-scaled inputs (= [0.1, 30]·sqrt(p) on [-1,1]); a regression test was added. The smoke tests showed test errors of all pool members on lshape_L1 (seed 0, N =

64, 256) and of the KRR and polynomial members on three problems; the pool, criteria, tie band, folds and all pre-registered thresholds were NOT changed after seeing them. Addendum v3.0 stated that no surrogate error on the new problems had been seen; this entry corrects that statement.

V10 SurroSelect pickling bug found by the plain-file validation (2026-11-04). After the `metric` option was added (v3.1), a fitted selector stored a lambda and could no longer be saved by `surroselect fit --model`; the command-line unit test had been run before the change and the full tool test suite was not re-run. The plain-file validation (experiment E8) exposed the failure ("Can't get local object"). Fix: the metric function is a property computed from the metric name. The registered evaluation (experiments C, S, R) uses the library interface without pickling and is unaffected; all tool tests were re-run after the fix.

S5. Data-leakage audit

Training, test and extrapolation inputs are drawn from separate named random streams. Over all 44 problems the number of exact input matches between the training pool and the test or extrapolation sets is 0. A planted-leak test (20 test samples copied into the training set of problem `heatcomp_L1`, $N = 64$) is flagged by the overlap check (20 exact matches, none without the leak), and the error on the leaked samples is 229 times lower than on the other test samples, showing that a leak would be visible. Validation errors do not depend on test labels (unit test, corrupting all test labels). Fitting steps (POD, scalers, hyper-parameters, cross-validation folds) use training runs only. Two properties remain: training subsets are nested within a seed, and seeds share the pool and test set, so seed variance understates sampling variance at large N. Comparisons in the mechanism study that use the best network by *test* error are oracle comparisons and are labelled as such.

S6. Per-problem results

Table S5 gives $\bar{y}$ (reduced basis against direct networks, validation-selected members, seed means) for every problem; positive values favour the reduced basis.

Table S3: $\bar{y}$ per problem and budget (positive: reduced basis better).

Problem	p	N=64	N=256	N=1024
advdiff_L1	4	-0.04	0.17	0.21
advdiff_L2	4	-0.09	-0.00	-0.07
bratu_L1	3	2.78	2.98	3.31
bratu_L2	3	2.14	2.97	3.08
burgers_L1	4	-0.06	0.68	2.25
burgers_L2	4	0.01	0.08	0.11
convdiff_L1	3	-0.03	0.60	0.48
convdiff_L2	4	-0.05	0.10	0.04
darcy_L1	4	0.15	1.12	1.13
darcy_L2	10	-0.00	-0.05	-0.19
darcyhd_L1	16	0.04	-0.02	-0.13
darcyhd_L2	32	0.03	0.00	-0.07
darcylin_L1	16	0.18	0.63	0.48
darcylin_L2	32	0.04	0.43	0.18
diff1d_L1	4	0.89	2.36	2.70
diff1d_L2	8	0.17	0.62	1.00
fisher_L1	4	-0.21	0.09	0.52
fisher_L2	4	-0.26	-0.26	-0.44
heat2d_L1	4	0.18	0.93	1.38
heat2d_L2	4	-0.04	-0.10	-0.34
heatcomp_L1	4	0.00	0.52	0.88
heatcomp_L2	9	-0.01	-0.06	-0.12
heatmove_L1	3	0.13	0.13	-0.17
heatmove_L2	5	-0.04	-0.02	-0.15
heatmv3_L1	3	0.12	-0.11	-0.44
heatp_L1	3	0.77	2.49	2.87
heatp_L2	3	1.08	2.71	2.90
helm_L1	3	1.20	2.69	2.93
helm_L2	3	-0.16	2.08	2.64
incl3_L1	3	0.02	0.05	-0.21
incl_L1	4	-0.02	-0.00	-0.25
incl_L2	4	0.02	-0.01	-0.13
kdv_L1	4	-0.14	-0.26	-0.23
kdv_L2	4	-0.04	-0.03	0.29
lshape_L1	4	0.20	1.14	1.66
lshape_L2	7	-0.11	0.01	-0.01
nlheat_L1	3	1.25	2.51	2.72
nlheat_L2	3	0.75	2.16	2.58
reacdiff_L1	3	0.85	1.62	2.57
reacdiff_L2	3	0.57	1.96	2.38
transport_L1	4	-0.26	0.05	0.75
transport_L2	4	-0.32	-0.23	-0.33
wave_L1	4	0.16	0.29	0.78
wave_L2	4	0.10	-0.06	-0.04

S7. The tool

Installation and use. The package `surroselect` (directory `surroselect/` of the repository; MIT licence, copyright holder to be set by the authors) depends on NumPy, SciPy and scikit-learn; PyTorch and Optuna are needed only for the optional `TunedMLP`. In Python:

```
sel = SurrogateSelector(cv=4).fit(X, Y)    # X: runs x inputs, Y: runs x outputs
sel.learning_curve()
print(make_report(sel, target=0.01))
Y_new = sel.predict(X_new)
```

On the command line, with plain `.npz` files that hold arrays `X` and `Y`:

```
surroselect fit train.npz --model m.pkl --target 0.01 --report report.md
surroselect predict m.pkl X_new.npy --out Y_pred.npy
surroselect evaluate m.pkl test.npz
```

Default pool. POD+GP (Matérn-5/2, rank 32), POD+GP (squared exponential, rank 32), POD+GP (Matérn-5/2, rank 8), POD+KRR with a wide grid of kernel, length-scale and ridge selected by closed-form leave-one-out error, POD+polynomial (degree up to six, at most 3000 features, leave-one-out), and a GP on the full field (Matérn-5/2 or squared exponential; only for $m \leq 2000$). An optional metric, the normalised RMSE $\|Y - \hat{Y}\|_F / \|Y - \bar{Y}_{\text{train}}\|_F$, is available for outputs with a large offset.

Validation of the tool on plain files. To check the tool itself, without the benchmark pipeline in the loop, three of the new problems (`heatcomp_L1`, `lshape_L1`, `convdiff_L1`; $N = 256$) were exported to `.npz` files and processed with the three command-line commands as separate processes. All 3 of 3 sequences ran without error, and the error printed by `evaluate` agreed with an independent computation from the saved predictions to within 10^{-12} in 3 of 3 cases (`results/s6/plain_files_validation.json`).

Tests. 9 unit tests for the tool (closed-form leave-one-out against brute force and its stability near interpolation, exact polynomial recovery, selection accuracy, the basis-limit flag, the power-law inversion, the report, the metric option and the command-line round trip), 6 tests for the finite-element problems and the 56 tests of the benchmark package are collected by the repository; `results/logs/tests_*.log` records the final runs. A fitted selector could not be saved in an early version after the metric option was added (deviation V10); the plain-file validation found it.

Known limits. The tool assumes exchangeable runs (A3); it does not model noise; for $p \gtrsim 10$ the default GP kernels have many hyper-parameters fitted on at most 300 runs; the forecast is an extrapolation without guarantee; and the optional neural member uses a small search budget.